

LOW STYRENE SBR: PATENT LANDSCAPE FOR COMPOSITION-CONTROLLED AND FUNCTIONALIZED STYRENE-BUTADIENE RUBBERS

1. ABSTRACT

This report surveys patents relevant to low styrene styrene-butadiene rubber (SBR), emphasizing materials in which styrene content is reduced relative to conventional SBR and/or tightly controlled through monomer feed, polymerization route, or post-polymerization functionalization. The selected patent set spans classic solution- and emulsion-polymerized SBR tread compositions, modern functionalized SBR architectures for silica-filled tire compounds, and broader conjugated-diene polymer platforms that explicitly define low vinyl-aromatic content windows. Across the landscape, the target is not simply “SBR” but low-styrene or styrene-lean SBR-like elastomers that preserve processability while balancing wet grip, rolling resistance, abrasion resistance, and hysteresis.

The patents show three dominant technical approaches. First, solution polymerization of styrene and butadiene with organolithium initiation is used to control styrene incorporation, vinyl content, molecular weight, and microstructure; this is seen in both older tread-rubber compositions and newer functionalized elastomers. Second, end-group or chain-mid functionalization is used to improve filler interaction, especially with silica, through silanol, silane, sulfur-silane, trithiocarbonate, or polyether-containing motifs. Third, some patents move beyond simple SBR to broader diene elastomers or polymer composites, but still disclose styrene-butadiene copolymer examples and composition ranges directly relevant to low-styrene SBR design.

Key formulation trends include silica reinforcement with silane coupling agents, use of aromatic or hydrocarbon oils, staged mixing to manage processability, and curing systems based on sulfur/accelerator packages. Process trends include batch and continuous solution polymerization, partial termination or coupling to create functional architectures, and post-polymerization modification at controlled temperatures. Overall, the evidence supports low-styrene SBR designs in the approximate 5-25 wt% styrene range, with solution polymerization, controlled vinyl content, and silica-compatible end-group functionality as the most relevant synthesis directions.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20260001972A1
- Patent Title: Conjugated diene polymer, rubber composition, and member for tire
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2026
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a conjugated diene polymer platform rather than a direct SBR synthesis example, but it explicitly defines a hydrogenated styrene-butylene copolymer containing a vinyl aromatic monomer fraction in a narrow low range. The polymer is positioned for tire rubber compositions where molecular architecture, entanglement, and controlled styrene-derived content are tuned to improve crack growth resistance and elongation at break. - Molecular weight between entangled portions: 2,400 g/mol or more and 3,000 g/mol or less — Claims 1 / Abstract - Mooney viscosity at 100° C.: 40 or more — Claims 1 / Abstract - Mooney stress relaxation (MSR): 0.8 or more — Claims 1 / Abstract - Vinyl aromatic monomer unit content: 1% by mass or more and 18% by mass or less — Claims 1 / Abstract - Weight average molecular weight: 300,000 or more — Claim 2 - Nitrogen content: 70 ppm or more and 300 ppm or less — Claim 3 - Modification ratio: 60% or more — Claim 3 - Mooney stress relaxation (MSR): 0.80 or more and 1.80 or less — Claim 4 - Difference alpha: 0.1 or less — Claim 5 - Nitrogen content: 90 ppm or more and 300 ppm or less — Claim 6 - Hydrogenation rate: 50 mol % or more — Claim 7 - Rubber component loading: 30 parts by mass or more of the conjugated diene polymer — Claim 8 - Natural rubber: 30 parts by mass or more — Claim 8 - Carbon black: 30 parts by mass or more — Claim 8

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance Highly relevant because it explicitly defines a low vinyl-aromatic content window up to 18 wt%, which overlaps the low-styrene SBR target space. Although broader than SBR, its composition/property constraints are directly useful for designing styrene-lean tire elastomers.

Patent 2

- Patent Number: US4866131A
- Patent Title: Rubber composition for high-performance tire tread
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 1989
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent teaches solution copolymerization of butadiene and an aromatic vinyl monomer in hydrocarbon solvent using an organic lithium initiator to make a low molecular weight copolymer, which is then blended with a second diene rubber. The disclosed tread composition uses the low molecular weight copolymer as a performance modifier and explicitly contrasts it with high-styrene SBR made by emulsion polymerization. - Low molecular weight copolymer content: 10 to 60% by weight — Abstract / Claim 1 - Molecular weight: about 2000 to about 50000 — Abstract / Claim 1 - Aromatic vinyl content X plus Y/2: not smaller than 25 and not larger than 70 — Claim 1 - Reaction medium: hydrocarbon solvent — Abstract / Claim 1 - Initiator: organic lithium compound — Abstract / Claim 1 - Polymerization temperature: 30°-80° C. — Example section - Age resistor: 0.7 g per 100 g copolymer — Example section - Press vulcanization: 160° C. for 25 minutes — Example 1 - Material rubber: 100 parts — Example 1 - Stearic acid: 1.5 parts — Example 1 - Zinc oxide: 3 parts — Example 1 - N-339 carbon black: 50 parts — Example 1 - Aromatic oil: 5 parts — Example 1 - Sulfur: 1.75 parts — Example 1 - Vulcanization accelerator: 1 part — Example 1 - Mixing temperature: 120° C. for 2 minutes — Example 5 - Silica: 25 parts — Example 5 - Silane coupling agent: 1.5 parts — Example 5 - Carbon black: 30 parts — Example 5 - Diethylene glycol: 2 parts — Example 5 - Open roll temperature: 50° C. — Example 5 - Press vulcanization: 160° C. for 30 minutes — Example 5

Relevant Experimental Evidence - Example 1: low molecular weight butadiene/aromatic vinyl copolymer was solution-polymerized and compounded into a tread rubber. - Example 4: varying the blend ratio of low molecular weight copolymer and high-molecular polymer changed abrasion, resiliency, and processability. - Example 5: silica-filled formulations showed improved abrasion resistance and resiliency with the invention blend. - Example 6: blending with natural rubber, emulsion SBR, or high-cis polybutadiene demonstrated compatibility across different diene rubbers. - Example 7: a branched and modified polymer also gave improved balance of properties versus the comparative blend.

Technical Relevance Very relevant because it directly addresses butadiene/styrene copolymer synthesis by solution polymerization and explicitly discusses high-styrene SBR as a comparator. The patent provides process and formulation guidance for tuning styrene-butadiene tread compounds toward lower styrene content and improved performance.

Patent 3

- Patent Number: US5902856A
- Patent Title: Rubber composition of two diene rubbers
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 1999
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution-state blending of a terminal-modified diene rubber with a high-molecular-weight branched diene rubber to obtain a processable rubber with improved abrasion resistance, resiliency, and low-temperature properties. The modified rubber is prepared from a diene polymer bearing active metal sites and reacted with a modifier, while the high-molecular component is obtained by coupling an active polymer with a polyfunctional coupling agent. - Modified diene rubber weight-average molecular weight: 150,000-500,000 — Abstract / Claim 1 - High-molecular diene rubber weight-average molecular weight: 800,000 or more — Abstract / Claim 1 - Modified rubber proportion: 5-80% by weight — Claim 1 - High-molecular rubber proportion: 95-20% by weight — Claim 1 - Vinyl bound content of conjugated diene units: 10-90% — Claim 1 - Conjugated diene content: at least 40% by weight — Claim 4 - Initiator amount: 0.65-0.3 mM per 100 g total monomers — Detailed description - Preferred initiator amount: 0.55-0.4 mM per 100 g total monomers — Detailed description - Polymerization solvent: hydrocarbon solvent — Detailed description - Example 1 mixing temperature: 60° C. for 30 minutes — Example 1 - Steam stripping coagulation — Example 1 - Vacuum drying: 60° C. for 24 hours — Example 1 - Press vulcanization: 160° C. for 25 minutes — Example 1 - Material rubber: 100 parts — Example 1 - Stearic acid: 1.5 parts — Example 1 - Zinc oxide: 3 parts — Example 1 - N-339 carbon black: 50 parts — Example 1 - Aromatic oil: 5 parts — Example 1 - Sulfur: 1.75 parts — Example 1 - Vulcanization accelerator: 1 part — Example 1 - Example 4 high-molecular polymer weight-average molecular weight: 1,200,000 — Table 5 - Example 5 silica loading: 25 parts — Table 6 - Example 5 silane coupling agent: 1.5 parts — Table 6 - Example 6 natural rubber: 20 or 40 parts — Table 7 - Example 6 emulsion SBR styrene content: 23.5 — Table 8 - Example 7 high-molecular polymer weight-average molecular weight: 1,150,000 — Table 10

Relevant Experimental Evidence - Example 1: solution blending of modified and high-molecular diene rubbers gave a processable rubber composition. - Example 2: changing the modifier still produced similar property improvements. - Example 3: different styrene/vinyl microstructures retained the invention effect. - Example 4: varying the blend ratio showed a synergy between processability and abrasion/resiliency. - Example 5: silica-filled compounds also benefited from the invention blend.

Technical Relevance Relevant because it provides a solution-state route for making diene rubbers with controlled microstructure and processability, including styrene-butadiene systems. While not a low-styrene SBR patent per se, it offers useful blending and molecular-weight control concepts for styrene-lean SBR formulations.

Patent 4

- Patent Number: EP3070120B1

- Patent Title: Polymer composite and method for producing the same, and rubber composition for ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a polymer composite for tire treads made by mixing a resinous organic compound with a high molecular weight polymer and a low molecular weight polymer, where the polymer is synthesized from a conjugated diene monomer and optionally an aromatic vinyl monomer. The emphasis is on pre-compositing the resin with the polymer to improve processability and balance grip and abrasion resistance. - Resinous organic compound molecular weight: 250 or more — Claims / Description - High molecular weight polymer: at least 50,000 — Claims / Description - Low molecular weight polymer: 3,000 or more and less than 50,000 — Claims / Description - Conjugated diene monomer examples: butadiene, isoprene, derivatives — Description - Aromatic vinyl monomer examples: styrene and derivatives — Description - Silica in tread examples: 80 phr — Example section - N234 carbon black: 1 phr — Example section - Aromatic oil: 39.5 phr — Example 3 - Silane coupling agent: 6.4 phr — Example 3 - ZnO: 2.5 phr — Example 3 - Stearic acid: 2 phr — Example 3 - Antioxidant: 1.9 phr — Example 3 - Anti-ozone wax: 1.5 phr — Example 3 - Diphenylguanidine: 1.5 phr — Example 3 - Sulfur: 1.2 phr — Example 3 - Sulfenamide: 2 phr — Example 3 - MES oil: 15 phr — Example 4 - Resin: 15 phr — Example 4 - Silica: 50 phr — Example 5 - Banbury fill level: 70% — Example 3/4/5 - Initial mixer temperature: about 90° C. — Example 3/4/5

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance Relevant because it explicitly includes styrene-butadiene-type polymers in the low molecular weight polymer class and focuses on tread performance tradeoffs central to low-styrene SBR. It is especially useful for formulation strategy, showing how resin/polymer pre-composites can improve processability without sacrificing grip and abrasion resistance.

Patent 5

- Patent Number: US8492479B2
- Patent Title: Process for preparing a diene copolymer comprising a polyether block, diene ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of styrene and butadiene in methylcyclohexane using n-butyllithium, followed by coupling with polyether-containing silane reagents to place polyether blocks in the middle of the SBR chain. The process is designed to improve filler interaction and maintain processing properties in silica-filled rubber compounds. - Nitrogen pressure: around 2 bar — Examples 1-4 - Solvent: methylcyclohexane — Examples 1-4 - Styrene charge: 156 g — Example 1 - Butadiene charge: 444 g — Example 1 - Tetrahydrofurfuryl ether: 0.45 mL — Example 1 - n-BuLi: 5.1 mmol — Example 1 - Polymerization temperature: 40° C. — Example 1 - Conversion rate: 95% — Example 1 - Initial inherent viscosity: 1.01 dL/g — Example 1 - Initial Mn: 82,000 g/mol — Example 1 - Hexamethylcyclotrisiloxane: 8.4 mL of 0.2 mol·L⁻¹ solution — Example 1 - Coupling temperature: 60° C. — Example 1 - Polyether silane: 11.8 mL of 0.22 mol·L⁻¹ solution — Example 1 - Coupling time: 90 min — Example 1 - Final inherent viscosity: 1.75 dL/g — Example 1 - ML viscosity: 55 — Example 1 - Final Mn: 178,000 g/mol — Example 1 - Styrene charge: 143 g — Example 2 - Butadiene charge: 407 g — Example 2 - Tetrahydrofurfuryl ether: 0.57 mL — Example 2 - n-BuLi: 5.1 mmol — Example 2 - Polymerization temperature: 40° C. — Example 2 - Conversion rate: 96% — Example 2 - Initial inherent viscosity: 0.96 dL/g — Example 2 - Initial Mn: 77,000 g/mol — Example 2 - Polyether silane: 19.6 mL of 0.14 mol·L⁻¹ solution — Example 2 - Coupling temperature: -15° C. — Example 2 - Coupling time: 90 min — Example 2 - Final inherent viscosity: 1.65 dL/g — Example 2 - ML viscosity: 54 — Example 2 - Final Mn: 167,000 g/mol — Example 2 - Styrene charge: 156 g — Example 3 - Butadiene charge: 444 g — Example 3 - Tetrahydrofurfuryl ether: 0.45 mL — Example 3 - n-BuLi: 4.5 mmol — Example 3 - Polymerization temperature: 40° C. — Example 3 - Conversion rate: 96% — Example 3 - Initial inherent viscosity: 1.00 dL/g — Example 3 - Initial Mn: 84,000 g/mol — Example 3 - Coupling temperature: 60° C. — Example 3 - Coupling time: 60 min — Example 3 - Final inherent viscosity: 1.61 dL/g — Example 3 - ML viscosity: 50 — Example 3 - Final Mn: 151,000 g/mol — Example 3 - Styrene charge: 1.93 kg — Example 4 - Butadiene charge: 5.21 kg — Example 4 - Tetrahydrofurfuryl ether: 675 mL of 0.065 mol·L⁻¹ solution — Example 4 - n-BuLi: 490 mL of 0.065 mol·L⁻¹ solution — Example 4 - Polymerization temperature: 45° C. — Example 4 - Conversion rate: 92% — Example 4 - Hexamethylcyclotrisiloxane: 635 mL of 0.0253 mol·L⁻¹ solution — Example 4 - Coupling temperature: 60° C. — Example 4 - ML viscosity: 58 — Example 4 - Mn: 161,000 g/mol — Example 4 - Trans-1,4 units: 20% — Example 4 - Cis-1,4 units: 19% — Example 4 - 1,2-units: 61% — Example 4 - Styrene content: 25% — Example 4 - (CH₃)₂Si functional groups: 3.73 mmol/kg — Example 4 - Styrene charge: 1.93 kg — Example 5 - Butadiene charge: 5.21 kg — Example 5 - Tetrahydrofurfuryl ether: 675 mL of 0.065 mol·L⁻¹ solution — Example 5 - n-BuLi: 505 mL of 0.065 mol·L⁻¹ solution — Example 5 - Polymerization temperature: 45° C. — Example 5 - Conversion rate: 92% — Example 5 - Inherent viscosity: 1.57 dL/g — Example 5 - ML viscosity: 54 — Example 5 - Mn: 158,000 g/mol — Example 5

Relevant Experimental Evidence - Example 1: mid-chain polyether-functionalized SBR was prepared by solution polymerization and coupling. - Example 2: a pendant polyether-block SBR was synthesized with similar solution conditions. - Example 3: a non-SiOH pendant polyether SBR was made, showing the coupling strategy is flexible. - Example 4: a larger-scale control SBR with

25% styrene and detailed microstructure was prepared. - Example 5: a non-functional SBR control was synthesized for comparison in silica-filled compounds.

Technical Relevance Highly relevant because it is a direct solution-polymerized SBR patent with explicit styrene/butadiene feed ratios and a 25 wt% styrene example, plus functionalization strategies that improve silica interaction. It provides a strong process template for low-styrene SBR synthesis and post-polymerization modification.

Patent 6

- Patent Number: EP2139703B1
- Patent Title: Coupled monomodal dienic elastomer having a silanol in chain medium function, ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses batch and continuous solution polymerization of styrene and butadiene using n-butyllithium, followed by coupling and mid-chain silanol functionalization via chlorosilane chemistry. The key innovation is a monomodal coupled SBR with a silanol function in the middle of the chain, intended for silica-filled tire tread compositions. - Batch polymerization solvent: cyclohexane 105 g (135 mL) — Example 1 - Butadiene/styrene mass ratios: 1 / 0.091 / 0.051 — Example 1 - Tetrahydrofurfuryl ethyl ether: 22 ppm — Example 1 - n-BuLi: 123 μ mol — Example 1 - Polymerization temperature: 40° C. — Example 1 - Conversion rate: 76% — Example 1 - Inherent viscosity: 0.94 dL/g — Example 1 - Mn: 92,000 g/mol — Example 1 - Continuous reactor volume: 32.5 L — Example 2 - Mass flows: cyclohexane/butadiene/styrene/tetrahydrofurfuryl ethyl ether = 100/11/3.2/0.045 — Example 2 - Line inlet n-BuLi: 160 μ mol per 100 g monomers — Example 2 - Reactor inlet n-BuLi: 1160 μ mol per 100 g monomers — Example 2 - Residence time: 40 min — Example 2 - Temperature: 70° C. — Example 2 - Conversion rate: 96% — Example 2 - Inherent viscosity: 1.28 dL/g — Example 2 - Mn: 83,000 g/mol — Example 2 - Methyltrichlorosilane coupling: 638 μ mol per 100 g monomers — Example 2 - Coupling time: 3 min — Example 2 - Water quench: 580 mmol per 100 g monomer — Example 2 - Antioxidant: 0.8 pce — Example 2 - Antioxidant: 0.2 pce — Example 2 - Tg: -38 °C — Example 3 - Silica: 80 pce — Example 3 - N234: 1 pce — Example 3 - Aromatic oil: 39.5 pce — Example 3 - Si69 coupling agent: 6.4 pce — Example 3 - ZnO: 2.5 pce — Example 3 - Stearic acid: 2 pce — Example 3 - Antioxidant: 1.9 pce — Example 3 - Anti-ozone wax: 1.5 pce — Example 3 - Diphenylguanidine: 1.5 pce — Example 3 - Sulfur: 1.2 pce — Example 3 - Sulfenamide: 2 pce — Example 3 - Tg: -25 °C — Example 4 - Silica: 80 pce — Example 4 - N234: 1 pce — Example 4 - MES oil: 15 pce — Example 4 - Resin: 15 pce — Example 4 - Si69 coupling agent: 6.4 pce — Example 4 - ZnO: 2.5 pce — Example 4 - Stearic acid: 2 pce — Example 4 - Antioxidant: 1.9 pce — Example 4 - Anti-ozone wax: 1.5 pce — Example 4 - Diphenylguanidine:

1.5 pce — Example 4 - Sulfur: 1.2 pce — Example 4 - Sulfenamide: 2 pce — Example 4 - Tg: -48 °C — Example 5 - Silica: 50 pce — Example 5 - N234: 1 pce — Example 5 - Si69 coupling agent: 4 pce — Example 5 - ZnO: 3 pce — Example 5 - Stearic acid: 1.9 pce — Example 5 - Antioxidant: 1.9 pce — Example 5 - Diphenylguanidine: 1 pce — Example 5 - Sulfur: 1.5 pce — Example 5 - Sulfenamide: 1.8 pce — Example 5 - Banbury fill level: 70% — Example 3/4/5 - Initial mixer temperature: about 90° C. — Example 3/4/5

Relevant Experimental Evidence - Example 1: batch SBR with mid-chain silanol functionality was prepared at 40 °C and 76% conversion. - Example 2: continuous SBR synthesis achieved 96% conversion at 70 °C with mid-chain coupling and silanol formation. - Example 3: silica-filled tread composition used the invention elastomer at Tg -38 °C. - Example 4: a higher-Tg SBR composition with resin and silica was formulated. - Example 5: a lower-Tg SBR composition with reduced silica loading was formulated.

Technical Relevance Directly relevant because it is a true SBR solution-polymerization patent with explicit styrene/butadiene feed ratios and silica-compatible mid-chain functionality. It supports low-styrene SBR design through controlled feed, continuous or batch operation, and post-coupling functionalization.

Patent 7

- Patent Number: US10676543B2
- Patent Title: Modified conjugated diene-based polymer, method for preparing the same and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of styrene and butadiene using a catalyst system based on an organic alkaline earth metal compound and an organic alkali metal compound to produce a high trans-1,4 styrene-butadiene random copolymer. The active polymer is then modified with an alkoxysilane-based modifier to improve silica interaction, rolling resistance, and wet-road performance. - Styrene charge: 19.7 g — Example 1 - Butadiene charge: 77.9 g — Example 1 - n-Hexane: 500 g — Example 1 - Barium di(ethylene glycol) ethyl ether: 1.3 mL of 30 wt% ethylbenzene solution — Example 1 - Trioctyl aluminum: 6.7 mL of 25 wt% hexane solution — Example 1 - n-Butyllithium: 1 mL of 2.5 M hexane solution — Example 1 - Polymerization temperature: 90° C. — Example 1 - Conversion rate: 90% — Example 1 - Modifier amount: 0.4 g — Example 1 - Modification temperature: 80° C. — Example 1 - Modification time: 30 minutes — Example 1 - Styrene charge: 20 g — Comparative Example 2 - Butadiene charge: 80 g — Comparative Example 2 - n-Hexane: 500 g — Comparative Example 2 - Tetramethylethylenediamine: 0.2 g — Comparative Example 2 - Polymerization temperature: 50°

C. — Comparative Example 2 - n-Butyllithium: 1 mL of 2.5 M hexane solution — Comparative Example 2 - Conversion rate: 98% — Comparative Example 2 - Modifier amount: 0.4 g — Comparative Example 2 - Modification temperature: 80° C. — Comparative Example 2 - Modification time: 10 minutes — Comparative Example 2 - Styrene charge: 19.7 g — Example 1 - Butadiene charge: 77.9 g — Example 1 - n-Butyllithium: 1 mL of 2.5 M hexane solution — Example 1 - Conversion rate: 90% — Example 1 - Styrene charge: 20 g — Example 2 - Butadiene charge: 80 g — Example 2 - Tetramethylethylenediamine: 0.2 g — Example 3 - Trans-1,4 bond content: 86 wt% — Experimental Example 1, Example 1 - Trans-1,4 bond content: 83 wt% — Experimental Example 1, Example 2 - Trans-1,4 bond content: 89 wt% — Experimental Example 1, Example 3 - Trans-1,4 bond content: 87 wt% — Experimental Example 1, Comparative Example 1 - Trans-1,4 bond content: 56 wt% — Experimental Example 1, Comparative Example 2 - Trans-1,4 bond content: 88 wt% — Experimental Example 1, Comparative Example 3 - Trans-1,4 bond content: 91 wt% — Experimental Example 1, Comparative Example 4 - Mw: 218,855 — Experimental Example 1, Example 1 - Mw: 211,143 — Experimental Example 1, Example 2 - Mw: 178,361 — Experimental Example 1, Example 3 - Mw: 221,593 — Experimental Example 1, Comparative Example 1 - Mw: 246,800 — Experimental Example 1, Comparative Example 2 - Mw: 204,278 — Experimental Example 1, Comparative Example 3 - Mw: 190,789 — Experimental Example 1, Comparative Example 4 - Mn: 113,987 — Experimental Example 1, Example 1 - Mn: 109,401 — Experimental Example 1, Example 2 - Mn: 87,432 — Experimental Example 1, Example 3 - Mn: 120,431 — Experimental Example 1, Comparative Example 1 - Mn: 154,329 — Experimental Example 1, Comparative Example 2 - Mn: 101,128 — Experimental Example 1, Comparative Example 3 - Mn: 98,345 — Experimental Example 1, Comparative Example 4 - Mw/Mn: 1.9 — Experimental Example 1, Example 1 - Mw/Mn: 1.9 — Experimental Example 1, Example 2 - Mw/Mn: 2.0 — Experimental Example 1, Example 3 - Mw/Mn: 1.8 — Experimental Example 1, Comparative Example 1 - Mw/Mn: 1.6 — Experimental Example 1, Comparative Example 2 - Mw/Mn: 2.0 — Experimental Example 1, Comparative Example 3 - Mw/Mn: 1.9 — Experimental Example 1, Comparative Example 4 - Modification ratio: 93% — Experimental Example 1, Example 1 - Modification ratio: 95% — Experimental Example 1, Example 2 - Modification ratio: 90% — Experimental Example 1, Example 3 - Modification ratio: 92% — Experimental Example 1, Comparative Example 2 - Modification ratio: 78% — Experimental Example 1, Comparative Example 3 - Modification ratio: 81% — Experimental Example 1, Comparative Example 4 - Mooney viscosity: 26 — Experimental Example 1, Example 1 - Mooney viscosity: 25 — Experimental Example 1, Example 2 - Mooney viscosity: 20 — Experimental Example 1, Example 3 - Mooney viscosity: 26 — Experimental Example 1, Comparative Example 1 - Mooney viscosity: 31 — Experimental Example 1, Comparative Example 2 - Mooney viscosity: 21 — Experimental Example 1, Comparative Example 3 - Mooney viscosity: 21 — Experimental Example 1, Comparative Example 4 - Silica: 70 parts by weight — Experimental Example 2 - Coupling agent: 11.2 parts by weight — Experimental Example 2 - Process oil: 5 parts by weight — Experimental Example 2 - Zinc oxide: 3 parts by weight — Experimental Example 2 - Stearic acid: 2 parts by weight — Experimental Example 2 - Antioxidant: 2 parts by weight — Experimental Example 2 - Anti-aging agent: 2 parts by weight — Experimental Example 2 - Wax: 1 part by weight — Experimental Example 2 - Rubber accelerator: 1.75 parts by weight — Experimental Example 2 - Sulfur: 1.5 parts by weight — Experimental Example 2 - Vulcanization accelerator: 2 parts by weight — Experimental Example 2 - Primary kneading temperature: 150°

C. — Experimental Example 2 - Primary discharge temperature: 145 to 155° C. — Experimental Example 2 - Secondary kneading temperature: 100° C. or less — Experimental Example 2 - Curing: 100° C. for 20 minutes — Experimental Example 2 - Abrasion test load: 10 N — Experimental Example 2 - Abrasion drum speed: 40 rpm — Experimental Example 2 - Abrasion test distance: 40 m — Experimental Example 2 - Dynamic mechanical analyzer frequency: 10 Hz — Experimental Example 2

Relevant Experimental Evidence - Example 1: high-trans random SBR was prepared and alkoxyisilane-modified. - Example 2: a TMEDA-containing variant was prepared for comparison. - Example 3: polar additive use altered the polymerization and modification behavior. - Experimental Example 1: examples showed high trans-1,4 content and high modification ratios. - Experimental Example 2: silica-filled compounds were prepared and tested for abrasion and viscoelasticity.

Technical Relevance Highly relevant because it is a direct styrene-butadiene random copolymer patent with explicit solution polymerization and post-modification, and it demonstrates how microstructure and functionalization affect tire-relevant properties. The styrene/butadiene feed and trans-1,4 control are especially useful for low-styrene SBR design.

Patent 8

- Patent Number: EP2596963B1
- Patent Title: Functionalized polymer, rubber composition with such a polymer and pneumatic ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses synthesis of functionalized SBRs using thiocarbonate or thioether end groups, followed by solution polymerization of styrene and butadiene with n-butyllithium and TMEDA. The resulting polymers are evaluated in silica-filled compounds to compare polymer-filler interaction, Payne effect, and cure behavior. - Mercaptopropyl trimethoxysilane: 30 mmol — Example 1 - Sodium methoxide: 30 mmol — Example 1 - Carbon disulfide: 40 mmol — Example 1 - Benzyl bromide: 30 mmol — Example 1 - Mercaptopropyl triethoxysilane: 30 mmol — Example 2 - Sodium ethoxide: 30 mmol — Example 2 - Styrene charge: 156 g — Example 5 - Butadiene charge: 444 g — Example 5 - Tetrahydrofurfuryl ether: 0.45 mL — Example 5 - TMEDA: 5.7 mmol — Example 5 - n-Butyl lithium: 1.6 mmol — Example 5 - Conversion rate: above 98% — Example 5 - Styrene-butadiene copolymer Mooney viscosity: 65 — Table 1, Sample 1 - Base Mn: 232,987 g/mol — Table 1, Sample 1 - Final Mn: not explicitly stated for Sample 1 in table excerpt — Table 1 - Styrene content: 17.97 wt% — Table 1, Sample 1 - Vinyl content: 48 — Table 1, Sample 1 - Tg: -25.0 °C — Table 1, Sample 1 - Mooney viscosity: 64.6 — Table 1, Sample 2 - Base Mn: 203,021 g/mol —

Table 1, Sample 2 - Final Mn: not explicitly stated for Sample 2 in table excerpt — Table 1, Sample 2 - Styrene content: 21.52 wt% — Table 1, Sample 2 - Vinyl content: 47.18 — Table 1, Sample 2 - Tg: -22.0 °C — Table 1, Sample 2 - Mooney viscosity: 77.6 — Table 1, Sample 3 - Base Mn: 198,883 g/mol — Table 1, Sample 3 - Final Mn: 226,667 g/mol — Table 1, Sample 3 - Styrene content: 21.94 wt% — Table 1, Sample 3 - Vinyl content: 46.78 — Table 1, Sample 3 - Tg: -23.8 °C — Table 1, Sample 3 - Mooney viscosity: 74.1 — Table 1, Sample 4 - Base Mn: 211,686 g/mol — Table 1, Sample 4 - Final Mn: 224,373 g/mol — Table 1, Sample 4 - Styrene content: 21.75 wt% — Table 1, Sample 4 - Vinyl content: 46.72 — Table 1, Sample 4 - Tg: -24.1 °C — Table 1, Sample 4 - Silica: 65 phr — Example 6 - Oil: 30 phr — Example 6 - Stearic acid: 2 phr — Example 6 - Zinc oxide: 3.5 phr — Example 6 - Sulfur: 1.7 phr — Example 6 - Sulfenamide: 1.6 phr — Example 6 - Diphenylguanidine: 1.5 phr — Example 6 - Antidegradant: 0.75 phr — Example 6 - Mixing: 4 min at 60 rpm, 140 °C starting temperature — Example 6 - Second non-productive mix: 3 min — Example 6 - Productive mix: 60 °C starting temperature, 60 rpm, 3 min — Example 6 - RPA cure: 160 °C, 7% strain — Example 6 - Static cure mimic: 160 °C for 30 min at 0.28% strain — Example 9/10

Relevant Experimental Evidence - Example 5: functionalized SBRs were synthesized and compared with non-functionalized controls. - Example 6: silica-filled compounds were prepared using a three-stage mix procedure. - Example 7: silane addition in the first non-productive stage reduced Payne effect for functionalized polymers. - Example 8: without silane, polymer-filler interaction differences became more pronounced. - Example 9: cured compounds with silane showed improved torque and filler interaction metrics. - Example 10: productive compounds without silane highlighted the role of functional end groups in silica interaction.

Technical Relevance Relevant because it directly addresses solution-polymerized SBR with styrene/butadiene feed and functional end groups that improve silica interaction. The disclosed styrene contents around 18-22 wt% are near the low-styrene boundary and provide useful comparative data for styrene-lean SBR design.

Patent 9

- Patent Number: EP2719549B1
- Patent Title: Functionalized polymer, rubber composition and pneumatic tire
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of styrene and butadiene in a batch reactor at 65 °C, followed by termination with isopropanol or cascade functional termination using ethylene sulfide and chlorotriethoxysilane. The resulting functionalized SBRs are then compounded with silica and evaluated for filler

interaction, cure behavior, and viscoelastic properties. - Reactor volume: 1 gallon — Example 1 - Polymerization temperature: 65 °C — Example 1 - Termination: isopropanol — Example 1 - Termination sequence: ethylene sulfide followed by chlorotriethoxysilane — Example 1 - GPC Mn: 250,000 g/mol — Table 2, Sample 1 - PDI: 1.42 — Table 2, Sample 1 - GPC Mn: 181,000 g/mol — Table 2, Sample 2 - PDI: 1.03 — Table 2, Sample 2 - GPC Mn: 197,000 g/mol — Table 2, Sample 3 - PDI: 1.10 — Table 2, Sample 3 - GPC Mn: 249,000 g/mol — Table 2, Sample 4 - PDI: 1.14 — Table 2, Sample 4 - GPC Mn: 206,000 g/mol — Table 2, Sample 5 - PDI: 1.18 — Table 2, Sample 5 - Mooney: 65 — Table 3, Sample 1 - Cis: 12.15 — Table 3, Sample 1 - Trans: 21.88 — Table 3, Sample 1 - Styrene: 17.97 — Table 3, Sample 1 - Vinyl: 48 — Table 3, Sample 1 - Tg: -25.0 °C — Table 3, Sample 1 - Mooney: 35.9 — Table 3, Sample 2 - Cis: 13.07 — Table 3, Sample 2 - Trans: 18.29 — Table 3, Sample 2 - Styrene: 23.70 — Table 3, Sample 2 - Vinyl: 44.9 — Table 3, Sample 2 - Tg: -22.08 °C — Table 3, Sample 2 - Mooney: 32.4 — Table 3, Sample 3 - Cis: 12.89 — Table 3, Sample 3 - Trans: 18.34 — Table 3, Sample 3 - Styrene: 23.46 — Table 3, Sample 3 - Vinyl: 45.31 — Table 3, Sample 3 - Tg: -22.09 °C — Table 3, Sample 3 - Mooney: 32.5 — Table 3, Sample 4 - Cis: 12.79 — Table 3, Sample 4 - Trans: 18.55 — Table 3, Sample 4 - Styrene: 23.73 — Table 3, Sample 4 - Vinyl: 44.93 — Table 3, Sample 4 - Tg: -22.00 °C — Table 3, Sample 4 - Silica: 65 phr — Example 2 - Oil: 30 phr — Example 2 - Stearic acid: 2 phr — Example 2 - Zinc oxide: 3.5 phr — Example 2 - Sulfur: 1.7 phr — Example 2 - Accelerators: 3.1 phr — Example 2 - Antidegradant: 0.75 phr — Example 2 - Mixing: 4 min at 60 rpm, 140 °C starting temperature — Example 2 - Second non-productive mix: 3 min — Example 2 - Productive mix: 60 °C starting temperature, 60 rpm, 3 min — Example 2 - RPA cure: 160 °C, 7% strain — Example 2 - Static cure mimic: 160 °C for 30 min at 0.28% strain — Example 5/6

Relevant Experimental Evidence - Example 1: functionalized SBR was synthesized by cascade termination with ES and CTES. - Example 2: silica-filled compounds were prepared using the functionalized SBRs. - Example 3: non-productive compounds with silane showed reduced strain dependency for functionalized polymers. - Example 4: without silane, functionalized polymers still improved filler interaction relative to control. - Example 5: productive compounds with silane showed distinct torque and Payne-effect behavior. - Example 6: productive compounds without silane highlighted the role of end-group chemistry in silica interaction.

Technical Relevance Relevant because it is a direct SBR functionalization patent with explicit styrene contents and solution polymerization conditions. The styrene levels around 18-24 wt% place the examples near the low-to-moderate SBR range and provide practical silica-compounding guidance.

Patent 10

- Patent Number: US7829624B2
- Patent Title: One-pot synthesis of nanoparticles and liquid polymer for rubber applications
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-pot synthesis in which a first monomer, optionally with a second monomer, is polymerized to a liquid polymer, partially terminated, and then further polymerized with mono-vinyl and multiple-vinyl aromatic monomers to form in situ nanoparticles. The resulting nanoparticle/liquid polymer blend is intended for rubber compositions, including tire tread applications, where improved dispersion and processing are desired. - Partial termination: 20 to 95% of polymers terminated — Claim 2 - Hydrocarbon solvent: used in the polymerization — Abstract / Claims - Pressure: sufficient to maintain a substantially liquid phase — Detailed description - Temperature range: about 20° C. to about 140° C. — Detailed description - Quenching/partial termination time: about 15 minutes — Detailed description - Crosslinking density Xd: 0.01 to 1 — Detailed description - Crosslinking density Xd: 0.1 to 0.8 — Detailed description - Crosslinking density Xd: 0.2 to 0.4 — Detailed description - Crosslinking density Xd: 0.3 — Detailed description - Crosslinking density Xd: 0.8 to 1.0 — Detailed description - Crosslinking density Xd: 0.9 — Detailed description - Organolithium initiator: preferred — Detailed description - Initiator amount: 0.01 to 1 phm — Detailed description - Initiator amount: 0.01 to 0.1 phm — Detailed description - Initiator amount: 0.025 to 0.07 phm — Detailed description - Mixer speed: 60 rpm — Example 4 - Master batch initial temperature: 110° C. — Example 4 - Final batch initial temperature: 75° C. — Example 4 - Tensile break stress: 16.14, 19.03, 20.76 MPa — Example 3 - Elongation at break: 483.4, 507, 452.4 % — Example 3 - 300% modulus: 8.03, 9.1, 11.61 — Example 3 - 50% modulus: 0.98, 1.12, 1.41 — Example 3 - Compound Tg: -45° C., -43° C., -40° C. — Example 3 - Rolling resistance tan delta at 50° C.: 0.207, 0.227, 0.202 — Example 3 - G' values: 3.779, 5.089, 5.880 — Example 3 - Tensile test temperature: 22° C. — Example 3 - Hysteresis test frequency: 1 Hz — Example 3 - Hysteresis test strain: 1% — Example 3 - Specimen geometry: ring width 0.05 inches, thickness 0.075 inches, gauge length 1.0 inch — Example 3 - Hysteresis specimen geometry: cylinder length 15 mm, diameter 10 mm — Example 3 - SBR 1: 100 phr — Example 4 - SBR 2: 90 phr — Example 4 - SBR 3: 90 phr — Example 4 - SBR 4: 90 phr — Example 4 - Carbon black N343: 50 phr — Example 4 - Aromatic oil: 15 phr or 0 phr — Example 4 - Zinc oxide: 3.0 phr — Example 4 - Hydrocarbon resin: 2.0 phr — Example 4 - Santoflex 13: 0.95 phr — Example 4 - Stearic acid: 2.0 phr — Example 4 - Wax: 1.0 phr — Example 4 - Nanoparticles: 0 or 10 phr — Example 4 - Liquid polymer/nanoparticle 2: 22.2 phr — Example 4 - Liquid polymer/nanoparticle 3: 25 phr — Example 4 - Sulfur: about 1.3 phr — Example 4 - Cyclohexylbenzothiazole sulfenamide: 1.4 phr — Example 4 - Diphenylguanidine: 0.20 phr — Example 4

Relevant Experimental Evidence - Example 3: compound properties showed how nanoparticle/liquid-polymer blends affect tensile and hysteresis behavior. - Example 4: formulations with SBR, carbon black, and nanoparticle/liquid-polymer blends were prepared and mixed under defined conditions.

Technical Relevance Indirectly relevant because it uses SBR in tire-compound formulations and addresses processability, dispersion, and rolling resistance. It is not a low-styrene SBR synthesis patent, but it provides useful compounding context for SBR-based tread systems.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Solution polymerization of styrene and butadiene with organolithium initiation is the dominant direct SBR synthesis route across the set (US4866131A, US8492479B2, EP2139703B1, US10676543B2, EP2596963B1, EP2719549B1).
- Explicit low-to-moderate styrene examples cluster around ~18-25 wt% styrene: EP2596963B1 reports 17.97 wt% in Sample 1 and 21.5-21.9 wt% in other samples; EP2719549B1 reports 17.97 wt% in the comparative functionalized SBR and 23.46-23.73 wt% in other samples; US8492479B2 Example 4 reports 25 wt% styrene.
- US4866131A uses a low molecular weight butadiene/aromatic vinyl copolymer blended with a high-styrene SBR, whereas the newer patents favor direct functionalization of solution SBR to improve silica interaction and tread performance.
- EP2139703B1 and US8492479B2 emphasize mid-chain or chain-end silanol/polyether functionality for silica-filled tread compounds, while US10676543B2 emphasizes high trans-1,4 microstructure plus alkoxy silane modification.
- US10676543B2 is the most microstructure-focused, explicitly requiring trans-1,4 bond content of 80 wt% or more; EP2596963B1 and EP2719549B1 also report trans/vinyl/styrene data but with lower styrene contents near the low-styrene boundary.
- Silica reinforcement and silane coupling are recurring formulation themes in the functionalized SBR patents (EP2139703B1, US10676543B2, EP2596963B1, EP2719549B1, EP3070120B1), indicating that low-styrene SBR is often optimized in silica-filled tread systems rather than carbon-black-only systems.
- Process conditions are generally mild to moderate solution-polymerization temperatures: 40-45 °C in US8492479B2 and EP2139703B1 batch examples, 65 °C in EP2719549B1, 70 °C in EP2139703B1 continuous example, and 80-90 °C in US10676543B2.
- Several patents rely on post-polymerization termination or coupling to introduce functionality: polyether-silane coupling in US8492479B2, methyltrichlorosilane coupling in EP2139703B1, alkoxy silane modification in US10676543B2, and ES/CTES termination in EP2719549B1.
- Older US4866131A focuses on blending a low molecular weight copolymer with a diene rubber, while later patents increasingly integrate the low-styrene SBR into a single functionalized polymer architecture to improve processability and filler interaction.
- US20260001972A1 broadens the landscape by defining a vinyl aromatic monomer content window of 1-18 wt%, which is directly compatible with low-styrene design targets even though the patent is not limited to SBR.

4. CONCLUSION

The evidence supports low-styrene SBR development primarily through solution polymerization of styrene and butadiene with tight control of monomer feed, microstructure, and end-group

functionality. The most practical synthesis window from the patent set is a styrene content near 18-25 wt% for many tire-grade examples, with lower values explicitly supported by the broader 1-18 wt% vinyl-aromatic window in US20260001972A1. For performance-balanced low-styrene SBR, the strongest evidence favors solution polymerization at about 40-70 °C, organolithium initiation, and silica-compatible functionalization such as silanol, polyether-silane, or alkoxysilane end groups. Where abrasion resistance is critical, high trans-1,4 microstructure and controlled modification ratio are beneficial. For tread compounding, silica plus silane coupling, sulfur/accelerator cure systems, and staged mixing are recurring best practices. Overall, the recommended synthesis direction is a solution-polymerized SBR with reduced styrene content, controlled vinyl/trans microstructure, and post-polymerization functionality designed to improve silica interaction and maintain processability.

5. REFERENCES

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- US5902856A | Rubber composition of two diene rubbers | Not disclosed | US | 1999 | <https://patents.google.com/patent/US5902856A/en>
- EP3070120B1 | Polymer composite and method for producing the same, and rubber composition for ... | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP3070120B1/en>
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- EP2139703B1 | Coupled monomodal dienic elastomer having a silanol in chain medium function, ... | Not disclosed | EP | 2018 | <https://patents.google.com/patent/EP2139703B1/en>
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